

Analysis supplement

MIDI-to-RMS Prediction: A Diagnostic Study Across Notes and Scales
16 September 2026

This supplied companion documents fixed-prediction analyses. The separate `analysis_companion/` folder contains arrays, fold membership, archived source, configurations and verification programs. A separate execution companion supplies the corrected training cache, retained model source, validation recipes and 12 seed-42 checkpoints. Recordings are not contained in these packages.

A. How often do trajectory and note gains agree?

Full and Note are the manuscript's primary, untransformed correlations. Every one of the 186 tracks enters each comparison below; there are no exact ties. The numerator counts tracks whose Full improves but Note decreases. Proportions and intervals are descriptive, conditional on these predictions. Whole piece/duplicate groups are resampled 5,000 times (seed 20260911), retaining all tracks and recomputing the numerator and denominator for each draw. These are not per-track significance tests or error rates for new data.

Mamba minus	Full up	Note down among these	Proportion	95% interval
Instrument OLS	152	50	32.9%	[24.4, 41.7]%
Shared envelope	155	23	14.8%	[8.9, 22.0]%

All four quadrants (Full up/Note up, Full up/Note down, Full down/Note up, Full down/Note down) contain 102/50/1/33 tracks against instrument OLS and 132/23/6/25 against the shared envelope. Thus most gains are concordant, while 51 and 29 tracks, respectively, have opposing Full and Note rankings. Full improves while log note-RMSE becomes worse on 51/152 and 75/155 tracks, respectively. These are separate criteria: a Note-correlation gain need not reduce numerical error.

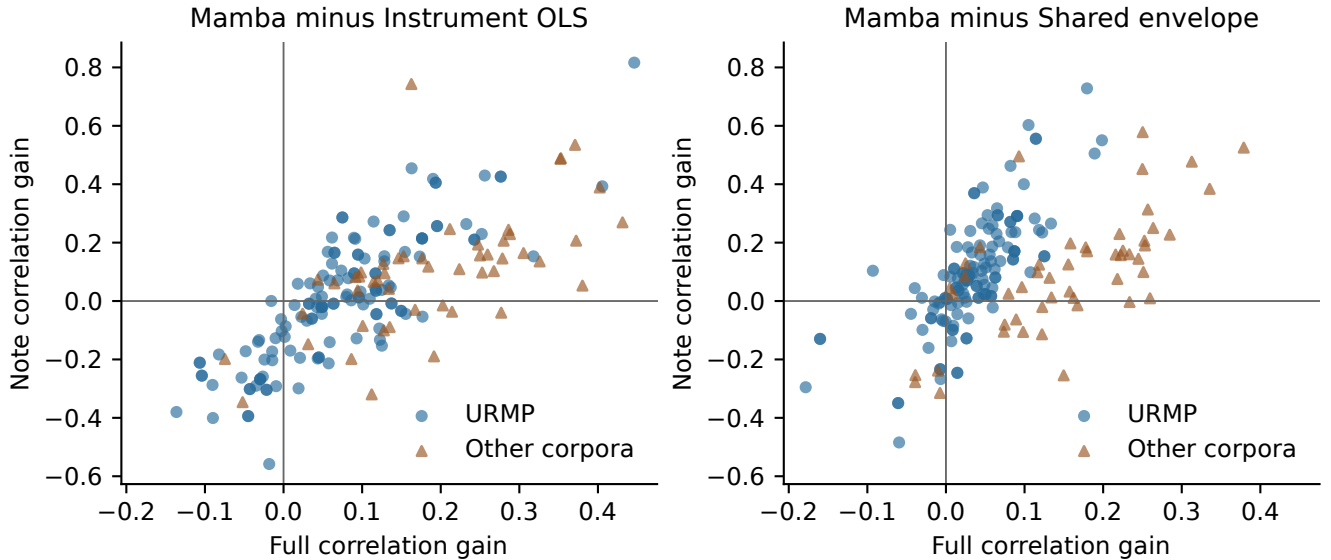


Figure S1. Every track is shown without jitter or filtering; both panels use identical limits. Marker and color distinguish URMP from the other corpora. Axes show whole-track paired differences, with zero marked.

B. Recorded-target and predicted RMS examples

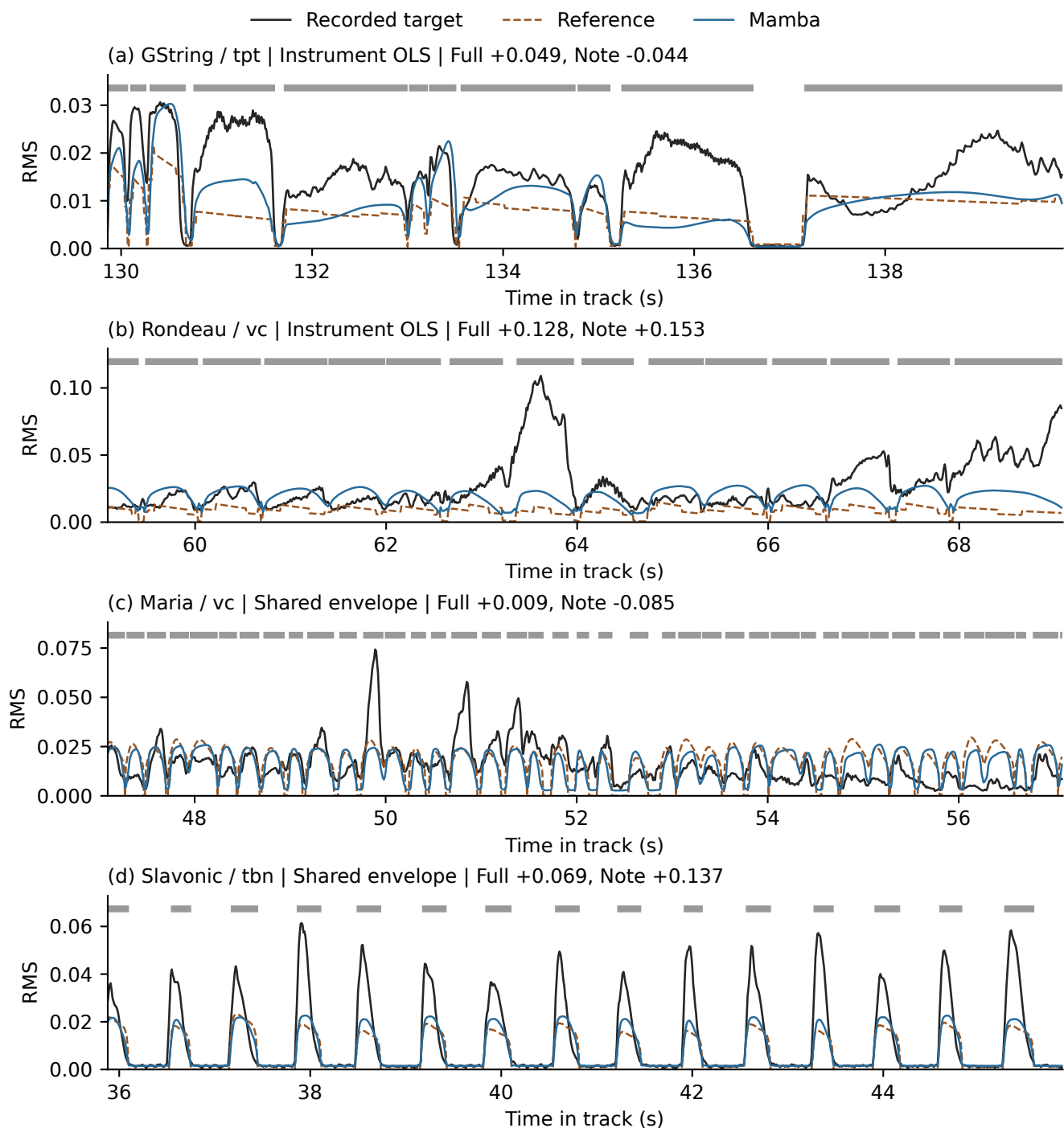


Figure S2. Unsmooth, unscaled RMS on the central ten seconds of each track. Gray bars are original note intervals. Panel titles report whole-track differences, not scores on the displayed window; the window need not reproduce the whole-track ordering. Vertical scales differ to show each recording's units. These are cached recorded RMS targets, not reconstructed recordings or listening evidence.

For each reference, one case from each Full-up/Note-down and Full-up/Note-up quadrant was selected by proximity to that quadrant’s median Full/Note gain, after IQR scaling of each coordinate. A zero IQR uses scale one; track identifiers break distance ties. Windows are always centered on the target horizon and are not selected by local error. This rule was recorded before computing this analysis, after earlier aggregate results had been reviewed; it is not preregistration. The cases illustrate both agreement and disagreement, rather than the largest effects.

C. Same-note interval sensitivity

All targets and predictions remain fixed. For each note, replace its end by the next strictly greater onset in the same track; simultaneous onsets share that end. At the final onset each note retains its original end. Use the original integer truncation at the stored hop, clip to the target horizon, and require at least four frames under both rules. The intersection retains all 30,352 original eligible notes. The alternative changes 19,796 intervals; Bach10 is unchanged because its cached offsets already follow that rule. No actual Bach10 note-offs are recovered.

The table gives Mamba minus reference Note correlation on the same notes, using equal track means and conditional 95% group-bootstrap intervals. Rule change is the paired difference of these gains (next-onset minus original). A change in whether an interval crosses zero is not itself evidence that the two gains differ.

Population	Reference	Original	Next onset	Rule change
All (186)	Instr. OLS	0.030 [-0.0047,0.068]	0.048 [0.0235,0.074]	0.017 [-0.0004,0.036]
All (186)	Shared	0.096 [0.0633,0.126]	0.071 [0.0480,0.092]	-0.026 [-0.0401,-0.009]
URMP (136)	Instr. OLS	0.005 [-0.0347,0.046]	0.027 [-0.0010,0.054]	0.022 [-0.0021,0.045]
URMP (136)	Shared	0.094 [0.0521,0.131]	0.056 [0.0279,0.080]	-0.038 [-0.0549,-0.019]
Other (50)	Instr. OLS	0.098 [0.0462,0.155]	0.104 [0.0566,0.157]	0.005 [-0.0080,0.020]
Other (50)	Shared	0.101 [0.0489,0.147]	0.110 [0.0655,0.152]	0.010 [0.0001,0.024]
Non-Bach (146)	Instr. OLS	0.014 [-0.0262,0.058]	0.036 [0.0091,0.065]	0.022 [-0.0005,0.044]
Non-Bach (146)	Shared	0.091 [0.0506,0.125]	0.059 [0.0327,0.081]	-0.033 [-0.0493,-0.012]

Extending to the next onset can include gaps and recorded tails, or shorten overlaps; it changes the estimand. Original cached offsets remain the primary analysis. The aggregate Mamba–instrument OLS Note interval becomes positive under the alternative rule. Its paired rule-change interval still includes zero. For the shared envelope, the gain stays positive but is smaller under the alternative. URMP/non-URMP differences persist descriptively under a common rule; they do not identify a causal corpus property or establish independent generalization.

Log note-RMSE for Mamba minus instrument OLS is -.056 [-.133,.023] originally and -.050 [-.125,.029] under next-onset intervals; both include zero. The complete per-track Note, Within and log note-RMSE values for all eight models, every population, and paired rule changes are in `refinement/results.json`.

D. Temporal context and verification

Original eligible note durations have quartiles .192/.290/.548 seconds (median .290); 72.0% are shorter than .5 seconds and 92.6% shorter than one second. These are pooled note summaries. The per-track unannotated-frame fraction has quartiles 7.7/19.0/30.1%; unannotated frames can contain recorded tails and are not necessarily acoustically silent. Gaussian sigma .5 or 1 second therefore measures variation across many of these notes. This can be appropriate for phrase dynamics, while not testing fine note-by-note control. No scale is chosen by score.

`refinement/ANALYSIS_PLAN.md` defines these added checks. `refinement/analyze.py` reproduces their results and figures. Original per-track Note and log note-RMSE match the frozen reports exactly; independent cumulative-sum interval means agree with direct interval means at numerical tolerance. The results record UTC, input hashes and code hashes. These checks preserve original predictions, folds and metrics. They add no training runs or independence from earlier researcher choices.

E. Full analysis methods

This companion defines the fixed-prediction analysis in the manuscript. It supplies cached targets, note annotations, MIDI features, predictions from all eight internal predictors, exact fold membership, analysis code and a file-hash manifest. It is a local submission package; no public repository or public data release is claimed. The supplied NumPy program reproduces the numerical comparisons without training or a GPU. Recordings are not included. The separate execution companion now supplies the corrected training cache, all twelve retained internal checkpoints and runtime records; it does not reconstruct the original recording pipeline.

1. Target and data provenance

There are 186 tracks: URMP 136, Bach10 40, PHENICX 2 and TRIOS 8. Piece identity and exact amplitude/pitch duplicates define 44 groups. `tracks.json` gives the original track identifiers, corpus, instrument, group, cached hop time and target hash. `folds.json` indexes this ordered track list, assigning training, validation and evaluation roles separately in each of four folds. Every track is evaluated once; no group crosses roles within a fold. Twenty percent of the remaining groups serve as validation. Split seed: 20260911.

Targets are the retained float32 amplitude arrays, without additional evaluation gain fitting or normalization. The archived preprocessing source specifies 16 kHz audio and `librosa.feature.rms` with a 320-sample window and 160-sample hop. That historical call leaves centering and padding at library defaults; its exact historical librosa version has not been recovered. The cached target arrays, rather than a claimed bit-exact reconstruction from recordings, define this study. They are not calibrated sound-pressure levels. Recording gain, room response, microphone placement and source separation can affect this target. A constant positive track gain does not change within-track Pearson correlation, but it does affect linear errors and may affect learning.

Archived loaders use instrument-part files: URMP `AuSep_*`, Bach10 instrument-suffixed WAVs, and instrument-named PHENICX/TRIOS files. They average channels to mono and resample to 16 kHz before RMS extraction; they do not select ensemble mixes. This is source-path evidence, not a new acoustic isolation measurement. Bach10 has onset annotations; its loader constructs each offset from the next onset, with the last extending to the retained horizon. All 1,917 adjacent Bach10 pairs in the cache match that rule. Other loaders retain supplied note durations. These conventions affect interval means and preclude attributing the URMP/non-URMP contrast solely to corpus acoustics. Notes contain absolute onset, offset, pitch and velocity. Velocity is zeroed before extracting every input feature, including features that might otherwise depend on it. No audio-derived velocity, teacher amplitude or audio feature is supplied at inference. Performed timing is supplied, so this is not score-to-performance generation. Note-interval RMS averages describe the observed track envelope; they do not identify isolated-note acoustic energy. Overlap exclusion below checks annotation interference but cannot eliminate reverberation or recording effects.

2. Input features and predictors

The 20 feature coordinates, in zero-based order, are: activity; pitch/127; normalized note phase; $\log(1 + \text{elapsed note time})$; onset proximity; offset proximity; local note density; preceding pitch interval; its absolute size; preceding rest; local pitch trend; rhythmic regularity; three instrument-family indicators; pitch position within track range; phrase position; relative duration; running pitch direction; articulation ratio. Context is computed from the supplied MIDI. Exact windows, clipping, overlap handling and boundary rules are in `source/local_run_20260912/source_snapshot/src/model/solo/dataset.py`, function `notes_to_frame_features`. The supplied feature arrays preserve the evaluated inputs.

Mamba uses model dimension 128, six layers, state size 16, convolution width 4, expansion 1, bidirectionality and dropout .2. S4D uses dimension 256, four layers, state size 64, bidirectionality and dropout .2. Both attach the same decoder design: a frame BiGRU, two Transformer layers with eight heads and feed-forward width 1024, and the existing note-aware amplitude head. Their total parameter counts are 7.66M and 8.59M. The independent standard BiGRU has two bidirectional layers, 256 units per direction and dropout .3; linear projections produce pitch logits and softplus amplitude, with 1.65M parameters. It was added after viewing the original results. This comparison is not an isolated ablation of the note-aware head.

Each neural run uses seed 42, batches of 32 random 512-frame crops, Adam without weight decay, initial learning rate .001, ten warm-up epochs, cosine decay to $1e-6$, at most 300 epochs and patience 60. Selection minimizes validation loss on eight

deterministic, equally spaced crops per track. Evaluation runs full tracks. Exact commands/configuration and model source are included under `source`; they document the frozen runs rather than a newly executed replication.

The loss is voiced-frame pitch cross-entropy plus twice the amplitude loss. With natural-log amplitude and epsilon $1e-7$, amplitude loss combines frame Huber loss (delta 1), voiced-frame correlation loss (weight .3), consecutive-voiced-frame first-difference MSE (.2), and stride-eight mean-pooled log-amplitude MSE (.3). The legacy `per_crop_amp_norm` flag subtracts the SAME target voiced-log mean from prediction and target; it does not fit separate prediction gain. Exact masking and empty-crop rules are in `compute_baseline_loss`. All other optional loss terms are disabled in the frozen recipes.

Five references use the same folds. The gate is one on the union of annotated note frames and zero elsewhere; it has no fitted level. Global OLS and instrument OLS fit an intercept and the 20 frame features to $\log(\max(\text{target}, 1e-7))$ on training frames. Instrument OLS fits one regression per instrument, with global fallback. Note OLS regresses $\log(\max(\text{note mean}, 1e-7))$ on mean note features using training notes of at least four frames. Exponentiated predictions use the fixed numerical log bounds $[-30, 10]$. No bound was active in the retained outputs. Shared envelopes multiply those note levels by instrument-specific templates averaging mean-normalized training notes in 64 phase bins. Templates are interpolated and renormalized at the requested duration. Rendered notes of at least one frame contribute; overlapping contributions are averaged and unannotated frames are zero. These references test timing, feature-based levels and reusable shapes, not the complete score-feature basis model of prior performance research.

3. Metrics, coverage and uncertainty

All reported means weight tracks equally. Pearson correlation is undefined with fewer than three samples or standard deviation below $1e-10$ in either array. Undefined values remain missing and their denominators are retained. Full-track correlation uses the complete cached horizon; active-frame correlation uses the union of annotated intervals.

For a note with times s, e and cached hop h , its frame interval is $[\max(0, \text{int}(s/h)), \min(T, \text{int}(e/h))]$. The original integer truncation and cached floating-point h are preserved. Eligible primary notes have at least four frames. Across-note correlation compares the predicted and target means of these intervals. Within-note shape first correlates frames within each eligible note, then averages defined note correlations within the track. These two measures are not additive components of the full-track coefficient. Note-wise affine invariance refers to each interval's correlation, not to a physically separable decomposition of an overlapping waveform.

For domain/scale evaluation, apply either identity or $\ln(a+1e-7)$, then Gaussian smoothing to both prediction and target. Standard deviations are 0, .05, .1, .2, .5 and 1 seconds, converted using the cached hop; boundary mode is reflect and truncation is four standard deviations. Zero smoothing is identity. No prediction is refitted for a measurement condition. The main result uses all scales; no best-scale model selection is performed.

In the frozen legacy report, `rmse_log` means full-track log RMSE and `rmse_log_active` means active-frame log RMSE. In the new display/export dictionary these are explicitly named `full_log_rmse` and `active_log_rmse`; original report fields are retained unchanged for provenance. Linear RMSE is $\sqrt{\text{mean}((\text{prediction}-\text{target})^2)}$; log RMSE replaces both arrays by $\ln(a+1e-7)$. Compute both on full tracks and active frames, then average tracks. No test-fitted gain or offset is used. Errors measure agreement in cached recording units, not perceptual quality. The main table uses log RMSE of note means; the four full/active frame-error views are supplied separately. Note OLS and shared envelopes have particularly large full-track log errors because their unannotated-frame predictions are zero while recorded tails need not be. The unit gate is an uncalibrated timing control, explaining its large error.

Paired intervals resample piece/duplicate groups with replacement, keeping every track of each sampled group, and recompute the equal-track mean difference. There are 5,000 draws, seed 20260911, with 2.5th/97.5th percentiles. Pair-specific defined coverage determines the group set. Intervals are pointwise and conditional on these fixed predictions, seed and folds; they do not measure training-seed variation, correct multiple comparisons or remove historical test-based development. They are descriptive precision checks for the studied predictors, not confirmatory tests of newly selected algorithms. A cross-zero interval neither establishes equivalence nor proves absence of improvement.

4. Annotation and population checks

The dated `PROTOCOL.md` records these additional analyses before their computation, after reviewing earlier results. This is not preregistration. Predictions, folds and targets are unchanged. All positive-length annotations contribute to overlap

detection, including notes shorter than the primary minimum. For the frame check, exclude an entire eligible note if any of its frames has multiplicity greater than one. For the independent continuous-time check, exclude both whole notes in every overlapping half-open time-interval pair, without quantization. No fragments are trimmed and no retained subset is chosen by scores.

Additional populations are whole tracks with no frame overlap; the previously frozen native single-lane population at the synthesis grid; exact-target deduplication; URMP; and all remaining corpora together. These definitions are distinct. All model means, coverage and planned paired comparisons are in `expected_results.json`, including unfavorable and positive exceptions. The small non-URMP subgroup is heterogeneous, so its contrast does not identify a causal corpus property.

5. Public synthesis systems

The public-system populations and frozen protocol JSON files are included separately. Neither public predictor uses MIDI velocity. The MIDI-DDSP expression input contains note pitch, note duration and instrument ID; original MIDI2Params receives frame pitches and onset/offset flags. Loudness/amplitude conditioning is generated by the respective predictor, not copied from the target recording. The supplied frozen protocol fields and inference adapters document these inputs. Original MIDI2Params uses the author-designated `best_model_cpu_120.pt` predictor and CustomViolin synthesis checkpoint on 46 violin tracks; it is not the MIDI-DDSP authors' reimplementation. Public MIDI-DDSP uses synthesis step 50000 and expression step 5000 on all 186 tracks. Both preserve absolute onsets on the 250 Hz synthesis grid, allocate overlapping notes to voices, and sum complete waveforms at unit gain. There is no score-based track trimming or gain selection. RMS is extracted from final 16 kHz audio, with window 320, hop 160, centered constant padding and the frozen target horizon. Seeds, tails, reverb and checkpoint hashes are in the supplied protocol files.

Public URMP pretraining may overlap the evaluation pieces. Neither pipeline was retrained on the internal folds. Public-system results describe fixed adapted pipelines and are not evidence of a fair architecture ranking or clean generalization. The companion also contains the already-collected final RMS arrays for all 186 MIDI-DDSP and 46 original MIDI2Params tracks, mapped to the same cached targets. Each source RMS file is checked against its original completion-record SHA-256. `verify_public_rms.py` uses NumPy/SciPy to recompute full, active, note-mean and within-note correlations plus log correlation and one-second log-smoothed correlation. It checks 1,392 per-track values, 36 public-subset means and 120 primary paired-bootstrap intervals against the frozen reports. These checks reproduce evaluation from archived final RMS, not waveform generation or RMS extraction from audio. No new prediction, gain fit, track selection or inference is performed.

6. Reproduction

Use Python with NumPy (verified with NumPy 2.4.3). From the extracted companion directory run `python verify.py`. It independently recomputes correlations, errors, overlap masks, 576 planned paired-bootstrap comparisons and fold group separation from the supplied arrays, checking against the stored results at tolerance $1e-8$. It uses no GPU and does not read remote files. The independent implementation uses difference-array overlap counting and sorted-interval overlap detection rather than the original audit loops. `verification.json` reports its actual result. The additional `verify_level_audit.py` independently checks 4,464 note-error entries and 45 paired error/aggregation intervals using cumulative-sum interval means and explicit resampled group-row concatenation; its maximum difference is $1.78e-15$. It does not import the original analysis implementation. `manifest.json` identifies the exact packaged files by SHA-256. Source files are supplied to clarify the original computation, including legacy options not enabled in this study; the frozen recipe files determine which options were used.

7. Low-variance interval check

`variance_audit.json` records target standard-deviation quantiles, per-track eligible/defined note-count distributions, and results for all eight predictors at thresholds $1e-8$, $1e-6$ and $1e-4$. No eligible target interval has standard deviation below $1e-6$; just one of 30,352 falls below $1e-4$. Every neural model and the shared envelope has defined shape correlations on all primary eligible notes. Track note counts range from 24 to 777 (median 140.5, quartiles 57.5 and 219). Note OLS has 569 defined correlations across 53 tracks. The stricter $1e-4$ target threshold changes neural shape means by less than $1.4e-5$.

Thresholds are in cached recording units, not perceptual thresholds. Arithmetic means of r describe a typical track; Fisher averaging or equal-note weighting would define a different estimand. Neither is silently substituted.

8. Note-level error and alternative aggregation

For each original eligible interval, average prediction and target separately. Note linear RMSE and MAE compare these means; note log RMSE compares $\ln(\text{mean}+1e-7)$, then takes a root mean square across notes. Finally average tracks equally. No test calibration is fitted. The main-table Error field is `note_log_rmse`, distinct from both `full_log_rmse` and `active_log_rmse`; their complete tables are retained. Correlation alone tests co-variation. Linear errors test numerical agreement in recording units; log differences are dimensionless predicted/target note-mean log ratios, with epsilon added before the ratio. They retain overall recording-gain mismatch because no gain is fitted.

The primary arithmetic mean of track correlations describes a typical instrument track without duration/note-count weighting. For sensitivity, average tracks inside each piece/duplicate group then average groups equally; alternatively clip track r to $\pm[1-1e-7]$, take arctanh , average equally over tracks, then apply $\tanh$. In each of 5,000 paired group-bootstrap draws (seed20260911), recompute the full estimator and difference. Fisher aggregation here changes the across-track aggregation, not the preceding arithmetic within-track note average. All planned comparisons and any interval-sign changes appear in Tables S5–S6 below and `level_and_aggregation_audit.json`. These exploratory alternatives do not replace the primary estimator. Run `python level_and_aggregation_audit.py` to reproduce them from the same arrays.

For the binary gate at zero smoothing, $\log(\text{gate}+\text{epsilon})$ is a positive affine map of the two gate values. Its Pearson change under the joint log transformation therefore comes from the target transformation. After smoothing, the scale comparison also changes temporal resolution. This is an explanatory property of the specified measurement, not an additional experiment.

F. Magnitude and active-frame scale checks

These checks describe the existing predictions retrospectively. Their plans, including their timing relative to prior results, are supplied with the arrays. **Table S3. Direction disagreement at numerical cutoffs.** For cutoff c , require Full gain $> c$ and Note loss $> c$ simultaneously. Each cell shows that count / tracks with Full gain $> c$; all 186 tracks are eligible. Cutoffs express absolute correlation-difference resolution, not perceptual importance, statistical significance or a preferred threshold. Every planned cutoff is shown.

Reference	$c = 0$	.01	.025	.05	.10
Instrument OLS	50/152	41/149	29/143	13/123	5/88
Shared envelope	23/155	15/144	9/124	6/91	2/47

Instrument OLS: within the original discordant quadrant, Full-gain quartiles are 0.036/0.061/0.127; Note-loss quartiles are 0.021/0.054/0.139.

Shared envelope: within the original discordant quadrant, Full-gain quartiles are 0.014/0.026/0.094; Note-loss quartiles are 0.021/0.085/0.121.

Table S4. Active-frame correlations, all six scales. Smooth each complete track before selecting the original active-frame mask, with reflect padding and truncation at four sigma. Log precedes smoothing; epsilon is 10^{-7} . This includes cross-boundary smoothing and is not a causal separation of note/gap effects. All defined cells cover 186 tracks; gate at zero smoothing is undefined (0 tracks).

Predictor	Domain	0	.05	.1	.2	.5	1 s
Gate	linear	–	0.228	0.251	0.284	0.327	0.353
Gate	log	–	0.363	0.411	0.467	0.533	0.571
Global OLS	linear	0.204	0.255	0.276	0.317	0.371	0.400
Global OLS	log	0.231	0.414	0.457	0.511	0.574	0.603
Instrument OLS	linear	0.284	0.323	0.341	0.377	0.432	0.461
Instrument OLS	log	0.308	0.443	0.480	0.531	0.600	0.638
Note OLS	linear	0.143	0.260	0.291	0.327	0.378	0.413
Note OLS	log	0.135	0.403	0.445	0.495	0.558	0.594
Shared envelope	linear	0.337	0.333	0.329	0.339	0.374	0.409
Shared envelope	log	0.403	0.448	0.473	0.507	0.556	0.590
Mamba	linear	0.447	0.442	0.445	0.466	0.501	0.520
Mamba	log	0.561	0.588	0.612	0.650	0.698	0.722
S4D	linear	0.430	0.425	0.432	0.457	0.500	0.521
S4D	log	0.534	0.565	0.595	0.637	0.690	0.717
BiGRU	linear	0.426	0.413	0.412	0.426	0.449	0.449
BiGRU	log	0.539	0.559	0.580	0.613	0.652	0.660

Mamba increases from .447 in linear amplitude to .561 in log amplitude without smoothing, and to .722 under one-second log smoothing. Thus the log-domain increase also appears when evaluation excludes unannotated frames. The smoothed gate becomes nonconstant within active frames because the full-track filter mixes boundaries; its positive active correlation does not indicate learned amplitude variation.

Epsilon sensitivity uses the complete grid $10^{-8}, 10^{-7}, 10^{-6}$ at all six scales. Largest absolute changes from the primary equal-track means are listed below; the underlying values and defined coverage are retained in `scale_results.json`.

Predictor	Full: max change	Active: max change
Gate	0.000324	0.000099
Global OLS	0.000328	0.000150
Instrument OLS	0.000320	0.000151
Note OLS	0.002879	0.003361
Shared envelope	0.006162	0.005095
Mamba	0.000332	0.000072
S4D	0.000332	0.000086
BiGRU	0.000338	0.000139

The larger shifts for Note OLS and shared envelopes occur in references with zero predictions outside notes; they are retained rather than conflated with the smaller neural-model changes. Higher smoothed scores can suit phrase-level evaluation. They do not, by themselves, test finer note-level control.

G. Error and aggregation tables

Table S5. Numerical errors of note means. Average means over each note first, then compute errors over notes and average tracks equally. No gain is fitted.

Predictor	Linear RMSE	Log RMSE	Linear MAE
Gate	0.963498	3.708426	0.963335
Global OLS	0.027222	0.814200	0.023670
Instrument OLS	0.026170	0.745229	0.022023
Note OLS	0.024914	0.723253	0.021389
Shared envelope	0.024916	0.723343	0.021391
Mamba	0.023161	0.688952	0.019876
S4D	0.024761	0.718275	0.021263
BiGRU	0.021582	0.649831	0.017626

Mamba minus	Error	Difference [95% interval]
Instrument OLS	Linear RMSE	-0.003 [-0.006230,0.000115]
Instrument OLS	Log RMSE	-0.056 [-0.133192,0.022958]
Instrument OLS	Linear MAE	-0.002 [-0.005260,0.000902]
Shared envelope	Linear RMSE	-0.002 [-0.004655,0.000852]
Shared envelope	Log RMSE	-0.034 [-0.106403,0.034737]
Shared envelope	Linear MAE	-0.002 [-0.004441,0.001121]
BiGRU	Linear RMSE	0.002 [-0.001146,0.004986]
BiGRU	Log RMSE	0.039 [-0.045899,0.140723]
BiGRU	Linear MAE	0.002 [-0.000509,0.005726]

Table S6. Aggregation sensitivity: Mamba minus reference. Track: arithmetic track mean. Group: arithmetic tracks within group, then equal groups. Fisher: transform each track r by $\arctanh$, average, then $\tanh$; recompute inside each draw. Values are gaps with conditional 95% group-bootstrap intervals. No estimator is selected by these results. Full error/overlap/population tables remain in `AUDIT_TABLES.md`.

Reference	View	Track	Group	Fisher
Instrument OLS	Full	0.106 [0.086,0.130]	0.113 [0.087,0.139]	0.103 [0.083,0.127]
Instrument OLS	Active	0.163 [0.136,0.194]	0.170 [0.139,0.203]	0.164 [0.134,0.197]
Instrument OLS	Note	0.030 [-0.005,0.068]	0.034 [-0.005,0.078]	0.017 [-0.020,0.058]
Instrument OLS	Within	0.272 [0.239,0.308]	0.276 [0.240,0.314]	0.276 [0.242,0.316]
Shared envelope	Full	0.068 [0.047,0.091]	0.068 [0.047,0.090]	0.064 [0.044,0.086]
Shared envelope	Active	0.110 [0.088,0.132]	0.110 [0.087,0.133]	0.113 [0.088,0.136]
Shared envelope	Note	0.096 [0.063,0.126]	0.083 [0.051,0.115]	0.092 [0.058,0.124]
Shared envelope	Within	0.075 [0.058,0.092]	0.069 [0.052,0.085]	0.075 [0.059,0.092]
BiGRU	Full	0.017 [0.002,0.034]	0.021 [0.008,0.036]	0.015 [0.000,0.030]
BiGRU	Active	0.021 [0.003,0.040]	0.027 [0.009,0.045]	0.021 [0.001,0.042]
BiGRU	Note	0.044 [0.006,0.081]	0.055 [0.020,0.090]	0.040 [-0.001,0.081]
BiGRU	Within	-0.003 [-0.019,0.010]	-0.007 [-0.018,0.003]	-0.003 [-0.019,0.010]

The Fisher Full interval for Mamba minus BiGRU has unrounded lower bound 0.00000477; its displayed zero is rounding, not strong evidence for stable model ranking. Fisher aggregation changes the Mamba–BiGRU Note interval to cross zero. It preserves interval signs for both primary Mamba/reference comparisons.

H. Unified directions for all three neural predictors

Table S7. Full gain, Note loss and error increase. All 186 tracks enter every comparison. Full up is the denominator for the next two counts. Note down means lower Note correlation; Error up means higher log note-RMSE. Medians concern the corresponding discordant subset, with no gain fitting. No exact ties occur. These are descriptive track directions, not per-track statistical tests.

Predictor	Reference	Full up	Note down	Error up	Median Note loss	Median error rise
Mamba	Instr. OLS	152	50	51	0.054	0.207
Mamba	Shared	155	23	75	0.085	0.096
S4D	Instr. OLS	158	52	57	0.071	0.135
S4D	Shared	152	30	55	0.078	0.137
BiGRU	Instr. OLS	156	72	75	0.096	0.147
BiGRU	Shared	138	35	59	0.100	0.190

Table S8. Conditional proportions [95% intervals]. Whole-group resampling, 44 groups, 5,000 draws, seed20260911; retain within-group tracks and recompute the ratio inside each draw. Intervals are conditional on the fixed predictions.

Predictor	Reference	Note down / Full up (%)	Error up / Full up (%)
Mamba	Instrument OLS	32.9 [24.4,41.7]	33.6 [23.7,44.9]
Mamba	Shared envelope	14.8 [8.9,22.0]	48.4 [39.2,58.5]
S4D	Instrument OLS	32.9 [23.2,42.5]	36.1 [25.9,47.1]
S4D	Shared envelope	19.7 [13.0,27.7]	36.2 [25.5,48.8]
BiGRU	Instrument OLS	46.2 [35.7,56.5]	48.1 [38.0,58.1]
BiGRU	Shared envelope	25.4 [16.7,36.2]	42.8 [29.3,57.3]

Table S9. Same numerical cutoff grid for every predictor. Each count is Full gain $> c$ and Note loss $> c$ / Full gain $> c$. These are absolute correlation differences, with no claim of perceptual significance. They do not threshold log errors.

Predictor	Reference	0	.01	.025	.05	.10
Mamba	Instrument OLS	50/152	41/149	29/143	13/123	5/88
Mamba	Shared envelope	23/155	15/144	9/124	6/91	2/47
S4D	Instrument OLS	52/158	46/155	31/143	13/122	3/81
S4D	Shared envelope	30/152	26/144	21/124	14/91	1/42
BiGRU	Instrument OLS	72/156	69/156	54/143	32/118	5/68
BiGRU	Shared envelope	35/138	24/124	18/110	13/81	3/46

All three predictors exhibit disagreement, but its frequency varies by predictor and reference. They share data and aspects of the training recipe, so this is not three independent replications. Complete quartiles and quadrant counts are recorded in `all_predictor_results.json`.

I. Whole-track note means for the same four examples

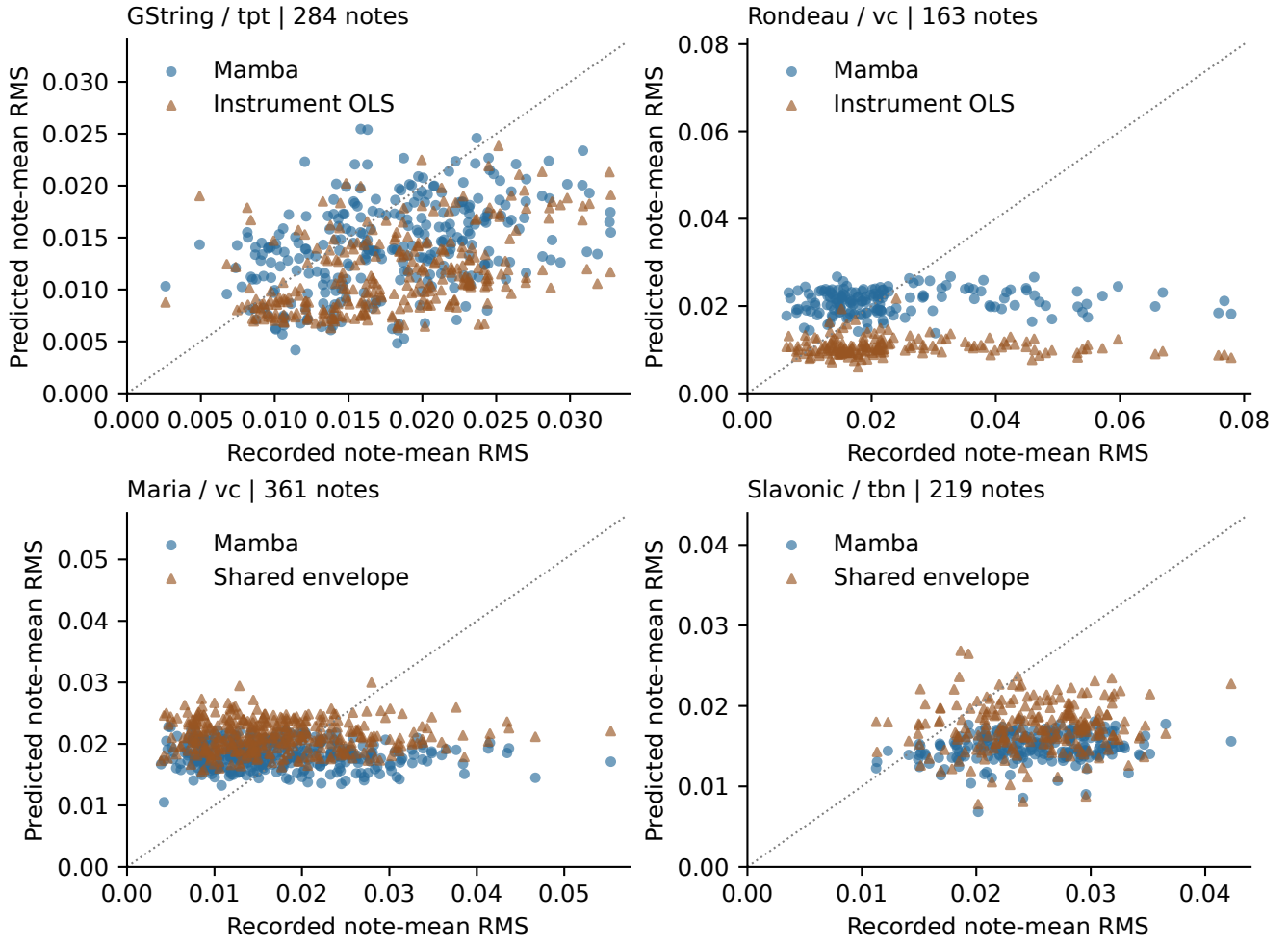


Figure S3. Each marker is one eligible original note interval from the entire track, including notes outside the ten-second windows in Fig. S2. Both predictors share the target x axis; each panel uses identical x/y limits and a dotted identity line, without fitted gain or offset. Axis ranges differ across recordings. The same four deterministic cases are used, so no further case selection is introduced. The plots show both co-variation and deviations from numerical agreement.

J. Dataset and fold composition

Table S10. Instrument tracks and corpus groups. Group counts below are computed from the supplied piece/duplicate labels. They are not counts of independent notes.

Instrument	Tracks	Corpus	Tracks	Groups
Bassoon	15	Bach10	40	10
Clarinet	21	PHENICX	2	2
Flute	18	TRIOS	8	5
Oboe	6	URMP	136	27
Saxophone	22			
Trombone	8			
Trumpet	23			
Viola	14			
Cello	13			
Violin	46			

Table S11. Fold sizes, tracks / groups. Training and validation roles are fold-specific; each track is evaluated exactly once. The original folds are unchanged.

Fold	Train	Validation	Evaluation
0	112/26	30/7	44/11
1	110/26	25/7	51/11
2	129/26	23/7	34/11
3	104/26	25/7	57/11

K. Adapted public-pipeline RMS metrics

Table S12. Public systems and matched Mamba predictions. Each row’s mean uses its stated tracks, with the original cached note rules. Public checkpoints were not retrained on the internal folds; MIDI-DDSP may have seen evaluation pieces. Different objectives and exposure preclude a controlled architecture ranking. Deduplication removes repeated exact targets; single lane is the frozen synthesis-grid population. Log/1s are full-track scores of unchanged final RMS, with log before smoothing. These columns are different measurement settings, not performance gains from training.

Pipeline	Population	Tracks	Full	Active	Note	Within	Log	Log/1s
MIDI-DDSP	All	186	0.523	0.329	0.211	0.432	0.677	0.757
Mamba	All	186	0.617	0.447	0.270	0.592	0.792	0.852
MIDI-DDSP	Dedup.	163	0.525	0.334	0.214	0.433	0.674	0.743
Mamba	Dedup.	163	0.623	0.458	0.282	0.594	0.794	0.847
MIDI-DDSP	Single lane	95	0.524	0.366	0.217	0.458	0.638	0.671
Mamba	Single lane	95	0.632	0.493	0.301	0.603	0.783	0.821
MIDI2Params	All	46	0.122	0.051	0.111	0.025	0.324	0.501
Mamba	All	46	0.510	0.374	0.232	0.535	0.728	0.841
MIDI2Params	Dedup.	37	0.134	0.063	0.138	0.028	0.326	0.477
Mamba	Dedup.	37	0.524	0.393	0.260	0.534	0.736	0.840
MIDI2Params	Single lane	18	0.158	0.106	0.240	0.029	0.293	0.391
Mamba	Single lane	18	0.558	0.460	0.317	0.565	0.753	0.831

The first Mamba block is matched to MIDI-DDSP’s population; the second to the 46-violin-track MIDI2Params population and its subsets. Mamba rows therefore refer to different track sets across blocks. Full per-track values, all available reference summaries and paired conditional intervals remain in `public_expected_results.json`.

L. Overall log offset and centered note error

For each track, let $d_n = \log(\hat{\mu}_n + \varepsilon) - \log(\mu_n + \varepsilon)$ on its N original eligible note intervals, with $\varepsilon = 10^{-7}$. Define $b = N^{-1} \sum_n d_n$, $c^2 = N^{-1} \sum_n (d_n - b)^2$, and $E^2 = N^{-1} \sum_n d_n^2$. Then $E^2 = b^2 + c^2$ exactly. Here b is the signed mean log error, c is the centered root mean square, and E is the original uncalibrated note log-RMSE. This identity partitions log-MSE, not correlation or the across-track mean RMSE.

Table S13. Equal-track summaries over all 186 tracks. Bars denote track averages; the last column is $\overline{b^2}/\overline{E^2}$. The mean-square components sum to $\overline{E^2}$, not $\bar{E}^2$.

Predictor	$\bar{b}$	$ \bar{b} $	$\bar{c}$	$\bar{E}$	$\overline{b^2}$	$\overline{c^2}$	Offset share
Gate	3.681	3.681	0.435	3.708	14.076	0.203	0.986
Global OLS	-0.350	0.583	0.448	0.814	0.599	0.216	0.735
Instrument OLS	-0.215	0.498	0.458	0.745	0.454	0.229	0.665
Note OLS	-0.213	0.491	0.437	0.723	0.429	0.206	0.675
Shared envelope	-0.213	0.491	0.437	0.723	0.429	0.206	0.675
Mamba	-0.151	0.451	0.425	0.689	0.397	0.195	0.671
S4D	-0.290	0.480	0.422	0.718	0.475	0.192	0.712
BiGRU	-0.255	0.404	0.445	0.650	0.282	0.211	0.573

These are descriptive components of the unchanged predictions. Subtracting b inside the error calculation does not create or evaluate a calibrated prediction. An overall log offset can have several causes; this calculation does not identify recording gain as its cause. Centered residuals likewise do not establish perceived dynamics or expressiveness.

Table S14. Among tracks where Full improves and uncalibrated Error increases, count increases in both squared components, only b^2 , or only c^2 . There are no cases where neither increases. Mean changes below use exactly that subset, comparing predictor minus reference; no new tracks are selected to improve an outcome.

Predictor	Reference	N	Both	b^2 only	c^2 only	$\Delta \overline{b^2}$	$\Delta \overline{c^2}$
Mamba	Instr. OLS	51	13	36	2	0.564	-0.024
Mamba	Shared	75	18	55	2	0.347	-0.017
S4D	Instr. OLS	57	15	42	0	0.609	-0.025
S4D	Shared	55	12	41	2	0.584	-0.019
BiGRU	Instr. OLS	75	27	46	2	0.290	-0.028
BiGRU	Shared	59	21	38	0	0.333	-0.018

For Mamba versus the shared envelope, 55 of the 75 Full-up/Error-up tracks have an increased squared mean log error without increased centered mean square; 18 have increases in both and two only in the centered component. Thus the increased uncalibrated error often includes a larger overall log offset. It should not all be interpreted as worse relative note relationships. The main Error metric remains unchanged.

This check was specified after reviewing the existing results, as recorded in `refinement/LOG_ERROR_PLAN.md`. The program `refinement/log_error_parts.py` recomputes every component from the original target and prediction arrays and records input and code hashes. Across 1,488 model-track pairs, E matches the archived error exactly; the maximum absolute residual of the decomposition identity is 1.32×10^{-14} . Complete per-track values are supplied in `refinement/log_error_parts.json`.

M. Minimum note length and Within correlation

This sensitivity check keeps original cached intervals, original predictions and the 10^{-10} standard-deviation threshold, and changes only the minimum interval length to 4, 10, 20 or 50 frames. At 100 Hz these correspond nominally to 40, 100, 200 and 500 ms. Eligible note counts are 30,352, 29,567, 22,347 and 8,629, respectively. Longer thresholds evaluate a different population; the four-frame condition remains primary. All planned thresholds are reported.

Table S15. Equal-track Within mean (defined notes / defined tracks). Gate has no defined Within correlations. Different coverage must not be treated as a common-population ranking, particularly for Note OLS.

Predictor	4 frames	10 frames	20 frames	50 frames
Gate	– (0/0)	– (0/0)	– (0/0)	– (0/0)
Global OLS	0.299 (30352/186)	0.301 (29567/186)	0.294 (22347/186)	0.296 (8629/181)
Instrument OLS	0.320 (30352/186)	0.326 (29567/186)	0.327 (22347/186)	0.332 (8629/181)
Note OLS	0.046 (569/53)	0.029 (552/53)	0.057 (430/45)	-0.061 (105/21)
Shared envelope	0.517 (30352/186)	0.520 (29567/186)	0.510 (22347/186)	0.468 (8629/181)
Mamba	0.592 (30352/186)	0.594 (29567/186)	0.584 (22347/186)	0.555 (8629/181)
S4D	0.581 (30352/186)	0.583 (29567/186)	0.568 (22347/186)	0.531 (8629/181)
BiGRU	0.595 (30352/186)	0.596 (29567/186)	0.585 (22347/186)	0.555 (8629/181)

Table S16. Paired Within differences and conditional 95% group-bootstrap intervals. At thresholds 4, 10 and 20 frames, all comparisons cover 186 tracks; at 50 frames they cover 181 tracks. All retain 44 groups. Each difference uses the common defined-track set for its predictor/reference pair, with 5,000 whole-group draws and seed 20260911. All six neural/reference intervals remain positive at every threshold; their sizes vary with the population.

Predictor minus instrument OLS

Predictor	4 frames	10 frames	20 frames	50 frames
Mamba	0.272 [0.239,0.308]	0.268 [0.235,0.304]	0.257 [0.226,0.292]	0.223 [0.192,0.250]
S4D	0.260 [0.229,0.298]	0.256 [0.225,0.292]	0.240 [0.205,0.278]	0.200 [0.163,0.233]
BiGRU	0.274 [0.243,0.309]	0.270 [0.239,0.304]	0.257 [0.228,0.290]	0.223 [0.191,0.251]

Predictor minus shared envelope

Predictor	4 frames	10 frames	20 frames	50 frames
Mamba	0.075 [0.058,0.092]	0.074 [0.057,0.091]	0.074 [0.053,0.093]	0.087 [0.058,0.114]
S4D	0.064 [0.048,0.080]	0.063 [0.047,0.079]	0.057 [0.032,0.079]	0.063 [0.037,0.089]
BiGRU	0.078 [0.063,0.095]	0.076 [0.061,0.093]	0.074 [0.058,0.092]	0.087 [0.059,0.113]

Mamba’s mean Within is .592/.594/.584/.555 as the minimum length increases; its gap over the shared envelope is .075/.074/.074/.087. Thus excluding very short notes does not remove the observed gain over this template. The 50-frame check excludes five whole tracks and most short notes; it does not establish that individual four-frame correlations are precise estimates.

The plan is `refinement/DURATION_PLAN.md`; the program and complete per-track results are `refinement/duration_check.py` and `refinement/duration_results.json`. This post-review check uses a centered dot-product calculation of Pearson correlation. The 1,488 original model–track comparisons match the archived four-frame results within 3.61×10^{-16} , including identical undefined-value coverage.

N. Corpus-specific paired comparisons

Table S17 reports the same primary seed-42 comparison on every corpus, plus the non-URMP aggregate. Counts are Note decrease / Full improvement; Error increase / Full improvement is shown separately. A track belongs to its recorded corpus; no result-based selection is used. PHENICX contains two tracks and is descriptive only. Group bootstrap intervals quantify conditional sampling uncertainty; they do not turn a small corpus into a broad validation set.

Corpus (tracks/groups)	Predictor	Instrument OLS		Shared envelope	
		Note down	Error up	Note down	Error up
URMP (136/27)	Mamba	38/104	26/104	14/109	49/109
URMP (136/27)	S4D	38/111	29/111	15/105	26/105
URMP (136/27)	BiGRU	49/112	59/112	17/97	48/97
Bach10 (40/10)	Mamba	10/39	21/39	6/37	21/37
Bach10 (40/10)	S4D	10/38	24/38	11/38	22/38
Bach10 (40/10)	BiGRU	20/34	10/34	16/33	5/33
PHENICX (2/2)	Mamba	1/2	1/2	1/2	2/2
PHENICX (2/2)	S4D	1/2	1/2	1/2	2/2
PHENICX (2/2)	BiGRU	0/2	1/2	0/2	2/2
TRIOS (8/5)	Mamba	1/7	3/7	2/7	3/7
TRIOS (8/5)	S4D	3/7	3/7	3/7	5/7
TRIOS (8/5)	BiGRU	3/8	5/8	2/6	4/6
nonURMP (50/17)	Mamba	12/48	25/48	9/46	26/46
nonURMP (50/17)	S4D	14/47	28/47	15/47	29/47
nonURMP (50/17)	BiGRU	23/44	16/44	18/41	11/41

Table S18. Mean paired changes in squared overall log offset (b^2) and centered log-error variance (c^2), across all 186 tracks. Their sum is the change in mean squared Error, not the difference of mean RMSE. No gain or offset is fitted to improve prediction. A smaller centered variance describes relative log-level agreement and does not establish perceptual improvement.

Predictor	Reference	Δb^2 [95% CI]	Δc^2 [95% CI]
Mamba	Instrument OLS	-0.057 [-0.191,0.079]	-0.034 [-0.048,-0.021]
Mamba	Shared envelope	-0.032 [-0.160,0.093]	-0.012 [-0.018,-0.005]
S4D	Instrument OLS	0.021 [-0.106,0.169]	-0.036 [-0.050,-0.023]
S4D	Shared envelope	0.046 [-0.064,0.176]	-0.014 [-0.024,-0.005]
BiGRU	Instrument OLS	-0.171 [-0.326,-0.037]	-0.018 [-0.035,0.001]
BiGRU	Shared envelope	-0.147 [-0.338,0.017]	0.004 [-0.008,0.019]

Complete corpus-specific mean differences, intervals, offset components and coverage are in `acceptance/corpus_results.json`. The program recomputes 4,650 values from the arrays independently of the earlier stored metrics, with maximum difference below 1.9×10^{-15} , and reconciles every full-population count with the main table.

O. Paired directions for fixed public outputs

Table S19. The same output diagnostic applied to both public pipelines against both references. Full-up denominators refer to the declared subset and reference; zero denominators mean no Full improvements, not a zero conditional error probability. These comparisons span different training protocols and possible piece exposure. They describe output disagreements and do not estimate fair architecture rankings or clean unseen-data generalization. All fixed protocols, checkpoints and adapters remain unchanged.

System / subset (tracks)	Reference	Full up	Note down	Error up
MIDI-DDSP / All (186)	Instrument OLS	97	16	38
MIDI-DDSP / All (186)	Shared envelope	68	10	37
MIDI-DDSP / Deduplicated (163)	Instrument OLS	84	15	34
MIDI-DDSP / Deduplicated (163)	Shared envelope	62	9	33
MIDI-DDSP / Single lane (95)	Instrument OLS	48	9	24
MIDI-DDSP / Single lane (95)	Shared envelope	38	6	25
MIDI2Params / All (46)	Instrument OLS	3	0	2
MIDI2Params / All (46)	Shared envelope	0	0	0
MIDI2Params / Deduplicated (37)	Instrument OLS	2	0	1
MIDI2Params / Deduplicated (37)	Shared envelope	0	0	0
MIDI2Params / Single lane (18)	Instrument OLS	1	0	0
MIDI2Params / Single lane (18)	Shared envelope	0	0	0

For MIDI2Params, Full exceeds instrument OLS on only 3 of 46 violin tracks, and never exceeds the shared envelope. This pipeline therefore provides few Full-improved cases for a conditional disagreement analysis. The complete output population is retained. Its fixed-output domain/scale changes remain a separate observation.

P. Execution assets and recovered source

The corrected training cache contains 186 tracks with original features, note annotations, RMS targets and auxiliary pitch labels. Its features, notes and RMS match the analysis companion exactly, and its four folds have identical membership. The execution companion includes all four best checkpoints for each internal model and their training histories/configurations. All 12 checkpoints load strictly; their saved full-fold predictions match the analysis arrays exactly. CPU inference on one deterministically selected shortest evaluation track per fold agrees at tolerance $\text{rtol } 10^{-3}$, $\text{atol } 10^{-5}$; this is not a claim of bitwise cross-device replay.

The recovered AAAI code archive contains preprocessing, model/training code, configurations and compact result summaries. Its 153/33 split, teacher conditions and old seeds belong to a different experiment and are not substituted for the present folds. The archive pins librosa 0.11.0, while the actual historical cache-creation environment is not independently established by that requirement file. Raw recordings and true Bach10 note-offs are not recovered by the archive. The main target remains the preserved arrays.

The execution README provides commands to rerun the retained internal recipe from this cache in a fresh output directory. Dataset licenses remain with their providers. This documentation supports reproduction of the cached-input experiment and metric analysis; it does not claim an independently repeated raw-audio preprocessing pipeline or an independent external validation.

Q. S4D training-seed repetitions

S4D was repeated with seeds 123 and 456 on the same four frozen folds and training recipe; seed 42 is the original retained run. Each seed has one complete outer prediction for each of the same 186 tracks and 44 groups. No seed is selected by its outer results and no track-by-seed replicas are treated as independent observations. Mamba and BiGRU results in preceding sections remain seed 42.

The new runs use the retained personal-device source unchanged, batch 32, 512-frame crops, the same losses, optimizer and fixed-window validation selection. All eight additional folds completed. Every role, prediction length, finite/nonnegative output, validation-selected epoch and source/data hash was checked. New repetitions use the retained recipe; historical seed-42 runtime/device migrations prevent an assertion of bitwise replay. These repetitions address training variability on these data, not historical researcher selection or independent new-piece validation.

Seed	Full	Active	Note	Within	Error
42	0.607	0.430	0.250	0.581	0.718
123	0.606	0.436	0.265	0.585	0.710
456	0.615	0.447	0.284	0.590	0.695

Table S20. Equal-track means. Error is log-RMSE of note means, not framewise training error. Each metric has 186 defined track values in every seed.

Seed	Reference	Full up	Note down	Error up	Median Note loss
42	Instrument OLS	158	52	57	0.071
42	Shared envelope	152	30	55	0.078
123	Instrument OLS	151	51	71	0.068
123	Shared envelope	142	26	64	0.047
456	Instrument OLS	163	51	67	0.064
456	Shared envelope	146	19	58	0.037

Table S21. Note-down and Error-up counts are conditional on Full improvement. Median Note loss is calculated on Full-up/Note-down tracks only. Against instrument OLS, the disagreement fractions are 32.9%, 33.8% and 31.3%. The mean Note gain also changes across seeds: the seed-456 interval is positive, so individual-track discordance does not establish absence of average note-level improvement.

Seed	Reference	Metric	Difference	Lower 95%	Upper 95%
42	Instrument OLS	Full	0.097	0.079	0.118
42	Instrument OLS	Active	0.146	0.123	0.170
42	Instrument OLS	Note	0.010	-0.027	0.050
42	Instrument OLS	Within	0.260	0.229	0.298
42	Instrument OLS	Error	-0.027	-0.098	0.053
42	Instrument OLS	b^2	0.021	-0.106	0.169
42	Instrument OLS	c^2	-0.036	-0.050	-0.023
42	Shared envelope	Full	0.058	0.040	0.079
42	Shared envelope	Active	0.093	0.070	0.115
42	Shared envelope	Note	0.076	0.034	0.112
42	Shared envelope	Within	0.064	0.048	0.080
42	Shared envelope	Error	-0.005	-0.063	0.062
42	Shared envelope	b^2	0.046	-0.064	0.176
42	Shared envelope	c^2	-0.014	-0.024	-0.005
123	Instrument OLS	Full	0.095	0.074	0.119
123	Instrument OLS	Active	0.152	0.124	0.181
123	Instrument OLS	Note	0.026	-0.012	0.066
123	Instrument OLS	Within	0.264	0.234	0.298

Seed	Reference	Metric	Difference	Lower 95%	Upper 95%
123	Instrument OLS	Error	-0.035	-0.115	0.046
123	Instrument OLS	b^2	-0.038	-0.184	0.114
123	Instrument OLS	c^2	-0.028	-0.042	-0.013
123	Shared envelope	Full	0.057	0.037	0.080
123	Shared envelope	Active	0.099	0.078	0.119
123	Shared envelope	Note	0.092	0.059	0.120
123	Shared envelope	Within	0.068	0.052	0.085
123	Shared envelope	Error	-0.013	-0.095	0.067
123	Shared envelope	b^2	-0.013	-0.162	0.133
123	Shared envelope	c^2	-0.005	-0.015	0.005
456	Instrument OLS	Full	0.105	0.085	0.128
456	Instrument OLS	Active	0.163	0.139	0.189
456	Instrument OLS	Note	0.044	0.015	0.075
456	Instrument OLS	Within	0.270	0.236	0.305
456	Instrument OLS	Error	-0.051	-0.127	0.035
456	Instrument OLS	b^2	-0.044	-0.193	0.118
456	Instrument OLS	c^2	-0.028	-0.043	-0.013
456	Shared envelope	Full	0.066	0.046	0.090
456	Shared envelope	Active	0.111	0.088	0.132
456	Shared envelope	Note	0.110	0.066	0.145
456	Shared envelope	Within	0.074	0.059	0.088
456	Shared envelope	Error	-0.029	-0.106	0.049
456	Shared envelope	b^2	-0.019	-0.164	0.131
456	Shared envelope	c^2	-0.006	-0.016	0.008

Table S22. Per-seed paired differences, S4D minus reference. Intervals resample the 44 piece/duplicate groups 5,000 times (random seed 20260911), with equal track weighting. They condition on the individual trained seed; the three intervals are not a combined seed-and-dataset uncertainty interval. All metric pairs here share 186 tracks. Full, Active, Note and Within prefer positive differences; Error, squared log offset b^2 and centered log-error variance c^2 prefer negative differences.

The stored per-track report includes conditional disagreement intervals and effect magnitudes for both references. An independent recomputation of the original seed-42 views agrees with the archived analysis to 3.7×10^{-15} ; all original target/feature/note arrays match the frozen training cache exactly. The accompanying result export preserves both new seeds' eight best checkpoints, predictions, validation histories, source and environments with SHA256 manifests.

R. Mamba repetitions and cross-seed interpretation

Mamba additionally uses seeds 123 and 456 on all four frozen folds, completing the planned sixteen new runs across Mamba and S4D. Each seed covers the same 186 tracks in 44 groups. Earlier tables retain seed 42; Section Q reports S4D and this section reports Mamba. All seeds are reported separately, without selecting one by its outer results or pooling repeated tracks as independent observations.

The eight new Mamba runs retain the source, data, features, folds, batch size, crops, losses, optimizer and validation selection specified before training. Seven runs used the local RTX 5070 Laptop with Python 3.12; the last seed-456 fold used Squishy’s RTX 3070 Ti Laptop with Python 3.13.7 and the same PyTorch 2.10.0+cu128. This previously unstarted fold was allocated after a local prelaunch memory check. Both use the 80% allocator limit. Source/data hashes and the unchanged last-fold protocol were verified. These are repetitions of the retained recipe across the recorded runtimes, not an isolated experiment varying only the random-number seed or a cross-device efficiency comparison.

Seed	Full	Active	Note	Within	Error
42	0.617	0.447	0.270	0.592	0.689
123	0.606	0.424	0.257	0.580	0.704
456	0.605	0.426	0.274	0.568	0.740

Table S23. Mamba equal-track means. Error is log-RMSE of note means. Every metric has 186 defined values per seed.

Seed	Reference	Full up	Note down	Error up	Median Note loss
42	Instrument OLS	152	50	51	0.054
42	Shared envelope	155	23	75	0.085
123	Instrument OLS	150	50	55	0.063
123	Shared envelope	149	28	66	0.080
456	Instrument OLS	152	47	66	0.081
456	Shared envelope	151	25	71	0.027

Table S24. Note-down and Error-up counts condition on Full improvement; their denominator is the Full-up column. Median Note loss uses the Full-up/Note-down tracks.

Seed	Reference	Metric	Difference	Lower 95%	Upper 95%
42	Instrument OLS	Full	0.106	0.086	0.130
42	Instrument OLS	Active	0.163	0.136	0.194
42	Instrument OLS	Note	0.030	-0.005	0.068
42	Instrument OLS	Within	0.272	0.239	0.308
42	Instrument OLS	Error	-0.056	-0.133	0.023
42	Instrument OLS	b^2	-0.057	-0.191	0.079
42	Instrument OLS	c^2	-0.034	-0.048	-0.021
42	Shared envelope	Full	0.068	0.047	0.091
42	Shared envelope	Active	0.110	0.088	0.132
42	Shared envelope	Note	0.096	0.063	0.126
42	Shared envelope	Within	0.075	0.058	0.092
42	Shared envelope	Error	-0.034	-0.106	0.035
42	Shared envelope	b^2	-0.032	-0.160	0.093
42	Shared envelope	c^2	-0.012	-0.018	-0.005
123	Instrument OLS	Full	0.095	0.075	0.119
123	Instrument OLS	Active	0.140	0.112	0.170
123	Instrument OLS	Note	0.018	-0.021	0.060
123	Instrument OLS	Within	0.259	0.225	0.298
123	Instrument OLS	Error	-0.041	-0.120	0.042
123	Instrument OLS	b^2	-0.039	-0.180	0.109

Seed	Reference	Metric	Difference	Lower 95%	Upper 95%
123	Instrument OLS	c^2	-0.032	-0.045	-0.019
123	Shared envelope	Full	0.057	0.037	0.080
123	Shared envelope	Active	0.087	0.064	0.109
123	Shared envelope	Note	0.083	0.050	0.113
123	Shared envelope	Within	0.063	0.046	0.081
123	Shared envelope	Error	-0.019	-0.093	0.056
123	Shared envelope	b^2	-0.015	-0.154	0.127
123	Shared envelope	c^2	-0.010	-0.017	-0.002
456	Instrument OLS	Full	0.095	0.075	0.118
456	Instrument OLS	Active	0.142	0.114	0.175
456	Instrument OLS	Note	0.035	-0.005	0.078
456	Instrument OLS	Within	0.248	0.219	0.281
456	Instrument OLS	Error	-0.005	-0.087	0.087
456	Instrument OLS	b^2	0.052	-0.108	0.235
456	Instrument OLS	c^2	-0.032	-0.046	-0.018
456	Shared envelope	Full	0.056	0.038	0.078
456	Shared envelope	Active	0.089	0.067	0.112
456	Shared envelope	Note	0.100	0.070	0.127
456	Shared envelope	Within	0.051	0.037	0.069
456	Shared envelope	Error	0.016	-0.055	0.095
456	Shared envelope	b^2	0.077	-0.065	0.239
456	Shared envelope	c^2	-0.010	-0.017	-0.002

Table S25. Mamba minus reference, per seed. Paired intervals resample the 44 groups 5,000 times (bootstrap seed 20260911). They condition on each trained seed and fixed folds; they are not combined seed-and-dataset intervals. Correlations favor positive differences; Error, squared log offset b^2 , and centered log-error variance c^2 favor negative differences.

Mamba shows Full-up/Note-down disagreement on 30.9–33.3% of Full-improved tracks against instrument OLS across seeds 42, 123 and 456. S4D shows Full-up/Note-down disagreement on 31.3–33.8% of Full-improved tracks against instrument OLS across seeds 42, 123 and 456. These trackwise disagreements can coexist with positive mean Note gains; they do not establish equivalence or absence of improvement. The complete paired intervals above and in Section Q show the dependence on reference and seed.

Every four-fold export passes unique coverage, disjoint roles, original feature/target/note timelines, finite checkpoints and predictions, and validation-selected epoch checks. The final-fold transfer is verified against a remotely generated SHA256 manifest. The Mamba export retains a separate union-completion record, original local paused queue, and last-fold allocation/runtime provenance. The accompanying analysis script reproduces all metrics and paired differences from the stored arrays. Repetitions assess variability on the existing works; they do not remove historical development exposure or replace independent new-work validation.

S. Frozen transfer to previously unused MusicNet works

Purpose and scope. The original 186-track study is retrospective. This additional experiment fixes inputs, predictions and analysis rules before examining outcomes on MusicNet. It tests transfer to solo-string recordings; it does not make the earlier model development prospective. The author confirmed no prior MusicNet use in ISMIR, AAAI or related development. Repertoire identities were checked against the previous corpora. The 21 recordings are movements of five Bach works, not 21 independent compositions. They are never pooled with the original 186 tracks.

Population. We include every official metadata row marked Solo Violin or Solo Cello: nine violin and twelve cello recordings. There is no exclusion by result, length, polyphony or amplitude. Solo playing can include chords. CSV labels aligned to the audio, rather than the unaligned reference MIDI, supply performed times. All labels and overlapping intervals are retained; interval means measure the solo recording during each interval, not the energy of an isolated note.

Work	Recordings	Release IDs	Labels
BWV1001	4	2241,2242,2243,2244	4370
BWV1002	3	2288,2289,2659	2281
BWV1006	2	2186,2191	2186
BWV1009	6	2217,2218,2219,2220,2221,2222	5340
BWV1010	6	2293,2294,2295,2296,2297,2298	5536

Target and features. The official raw release archive was checksum-verified. Its MD5 is 844764911fa0d5b97c97da944a057590; individual WAV/CSV SHA256 values are recorded in the dataset manifest. All source WAVs are mono, 44.1 kHz. Float32 PCM is resampled to 16 kHz with librosa 0.11.0, `soxr_hq`. RMS explicitly uses a 320-sample window, 160-sample hop, centering and constant zero padding. No gain normalization, trimming, tail removal or target-dependent feature enters the input. Unknown historical cache padding is not retrospectively inferred from this new pipeline.

CSV times are original 44.1-kHz sample indices. Onsets and offsets are divided by the source rate, then cast to float32 for the frozen feature extractor. Velocity is zero. The original 20-feature extractor and its onset-order/overlap overwrite behavior are preserved; recording instrument is set by release metadata. Some violin CSVs use several string-family MIDI programs (41/42/43). They are retained, not used to exclude voices. These are documented annotation properties, not evidence of three separate physical performers. The float32 hop is 0.009999999776482582 seconds. Metric interval endpoints use integer truncation at that hop and clipping only to the complete recording horizon.

There are 19,713 labels, including 2,685 overlapping-note entries under the audit definition. The largest time-cast discrepancy is 15.25 microseconds. The per-recording audit retains source lengths, first onset, last offset and full waveform duration.

Frozen predictors. The first numbered original fold (fold 0) and seed 42 were fixed before new outcomes. Mamba/S4D best checkpoints retain their original fixed-window validation-loss selection. Global, instrument and note OLS coefficients and shared templates come only from that fold’s original training role. No MusicNet retraining, fine-tuning, calibration or model search occurs. BiGRU and public synthesis pipelines are not additional candidates in this transfer study.

Model-choice illustration. All 30 original fold-0 validation tracks are evaluated as full sequences for five fixed candidates. Highest mean Full selects Mamba; highest mean Note selects S4D. These selections were saved before MusicNet inference. The illustration compares fixed predictors; it does not establish that one metric is universally the better selection rule.

Candidate	Validation Full	Validation Note
Instrument OLS	0.531	0.247
Mamba	0.679	0.304
Note OLS	0.458	0.259
Shared envelope	0.585	0.259
S4D	0.663	0.338

Runtime verification. Both networks use their unchanged float32 state dictionaries, evaluation mode, complete sequences and batch one. Runtime-only attention backend changes were checked against a CPU float32 reference before new predictions (maximum absolute difference 3.73×10^{-8} on the check). TF32 is disabled; there is no context truncation or half-precision inference. A 34,000-frame feasibility input was made from original training features. Local RTX 5070 ran Mamba and Squishy RTX 3070 Ti ran S4D, each with an allocator limit of 80%. Both completed 21/21 recordings, with peak allocated

memory approximately 1.00/1.03 GiB. Runtime logs describe their respective environments; timings are not a cross-hardware efficiency comparison. Failed pre-inference setup checks and their corrections are preserved separately from completed runs.

Metrics and uncertainty. The same Full, Active, Note, Within, note log-RMSE Error, squared mean log error b^2 and centered error variance c^2 are used. Note eligibility is at least four frames; correlation requires at least three samples and both standard deviations above 10^{-10} . Undefined values remain undefined. The complete 147 model–recording pairs are checked for unique IDs, exact horizon, finite nonnegative predictions and the identity $E^2 = b^2 + c^2$. An independent error recomputation agrees within 8.9×10^{-16} .

Equal-recording summaries retain all available values; equal-work summaries first average within each composition. Paired intervals below are exploratory 95% percentiles from 5,000 draws of five whole works, using NumPy default RNG seed 20260916 and retaining all recordings of each selected work. Five groups provide limited population information. These are not 21-independent-sample or ten-instrument confirmatory intervals.

T. Transfer results, coverage and work-specific effects

All point means and all fixed predictors are reported, including the timing gate. Lower Error, b^2 and c^2 indicate better numerical agreement. Constant-prediction correlations are undefined, not zero.

Predictor	Full	Active	Note	Within	Error	b^2	c^2
Gate	0.162	–	–	–	3.125	9.628	0.281
Global OLS	0.144	0.063	0.064	0.049	1.129	1.095	0.291
Instrument OLS	0.076	0.002	-0.056	0.023	0.959	0.647	0.387
Note OLS	0.147	0.046	0.059	0.101	1.069	0.941	0.297
Shared envelope	0.141	0.059	0.062	0.026	1.069	0.941	0.296
Mamba	0.176	0.115	0.102	0.066	0.876	0.574	0.297
S4D	0.196	0.129	0.128	0.083	0.992	0.790	0.278

Coverage (recordings with defined values out of 21).

Predictor	Full	Active	Note	Within	Error	b2	c2
Gate	21	0	0	0	21	21	21
Global OLS	21	21	21	21	21	21	21
Instrument OLS	21	21	21	21	21	21	21
Note OLS	21	21	21	14	21	21	21
Shared envelope	21	21	21	21	21	21	21
Mamba	21	21	21	21	21	21	21
S4D	21	21	21	21	21	21	21

Equal-work point means (five composition means).

Predictor	Full	Active	Note	Within	Error	b^2	c^2
Gate	0.169	–	–	–	3.161	9.867	0.285
Global OLS	0.177	0.099	0.133	0.048	1.090	1.014	0.286
Instrument OLS	0.084	-0.012	-0.105	0.044	1.022	0.800	0.371
Note OLS	0.155	0.054	0.068	0.086	1.107	1.026	0.310
Shared envelope	0.151	0.064	0.070	0.029	1.107	1.026	0.310
Mamba	0.169	0.094	0.072	0.058	0.950	0.723	0.305
S4D	0.186	0.105	0.085	0.075	1.011	0.833	0.287

Full and Note are substantially lower than on the original study. Against instrument OLS, Mamba has 3/19 and S4D 2/21 Full-up/Note-down recordings; their mean Full and Note gains are positive in each of the five works. This does not reproduce the original 31–34% discordance rate. The shared-envelope Full advantage over Note OLS does not replicate: its new mean difference is -0.0058. A shared-shape control must therefore be tested on the target population rather than assumed to improve every corpus.

The selection rules choose different predictors. S4D minus Mamba has mean Full +0.0197, Note +0.0261, Error +0.1152 and centered variance -0.0187. The five-work Note and Error intervals include zero; centered variance improves in four works. These point summaries distinguish relative-level from overall-level agreement, but do not demonstrate a reliable universal benefit of Note-based selection. Equal-work Error difference is +0.0618, showing the influence of work weighting.

Mamba versus the shared envelope also illustrates why the error decomposition is necessary: on MusicNet its Error falls by 0.1928 while centered variance changes by +0.0004. This improvement is chiefly an overall-offset change, unlike the clear centered-error reduction in the original sample. No acoustic cause is identified by the algebra alone.

Comparison	Full up	Note down among these
Mamba – Instrument OLS	19	3
Mamba – Shared envelope	16	4
S4D – Instrument OLS	21	2
S4D – Shared envelope	17	3
Shared envelope – Note OLS	7	4
S4D – Mamba	15	4

Complete paired contrasts. Each row uses matched recordings. Intervals resample whole works. The final column averages the five work effects equally. **Mamba minus Instrument OLS**

Measure	Recordings	Mean difference	Exploratory interval	Equal-work difference
Full	21	0.100	[0.043, 0.135]	0.084
Active	21	0.113	[0.072, 0.142]	0.107
Note	21	0.158	[0.116, 0.225]	0.177
Within	21	0.043	[-0.045, 0.094]	0.014
Error	21	-0.082	[-0.139, -0.028]	-0.073
b2	21	-0.073	[-0.165, 0.035]	-0.077
c2	21	-0.090	[-0.143, -0.003]	-0.066

Mamba minus Shared envelope

Measure	Recordings	Mean difference	Exploratory interval	Equal-work difference
Full	21	0.035	[-0.026, 0.065]	0.018
Active	21	0.056	[-0.030, 0.099]	0.031
Note	21	0.040	[-0.101, 0.113]	0.002
Within	21	0.040	[0.010, 0.055]	0.029
Error	21	-0.193	[-0.257, -0.077]	-0.157
b2	21	-0.367	[-0.500, -0.164]	-0.303
c2	21	0.000	[-0.031, 0.024]	-0.005

S4D minus Instrument OLS

Measure	Recordings	Mean difference	Exploratory interval	Equal-work difference
Full	21	0.120	[0.059, 0.151]	0.102
Active	21	0.128	[0.089, 0.148]	0.118
Note	21	0.184	[0.120, 0.255]	0.190
Within	21	0.060	[-0.037, 0.119]	0.030
Error	21	0.033	[-0.130, 0.140]	-0.011
b2	21	0.143	[-0.252, 0.375]	0.032
c2	21	-0.109	[-0.160, -0.024]	-0.084

S4D minus Shared envelope

Measure	Recordings	Mean difference	Exploratory interval	Equal-work difference
Full	21	0.055	[-0.009, 0.091]	0.035
Active	21	0.071	[-0.021, 0.119]	0.042
Note	21	0.066	[-0.116, 0.157]	0.015
Within	21	0.057	[0.020, 0.080]	0.045
Error	21	-0.078	[-0.154, -0.031]	-0.095
b2	21	-0.151	[-0.333, -0.047]	-0.193
c2	21	-0.018	[-0.042, 0.003]	-0.023

Shared envelope minus Note OLS

Measure	Recordings	Mean difference	Exploratory interval	Equal-work difference
Full	21	-0.006	[-0.013, 0.009]	-0.004
Active	21	0.012	[-0.004, 0.023]	0.009
Note	21	0.002	[0.000, 0.004]	0.002
Within	14	-0.100	[-0.189, -0.034]	-0.092
Error	21	0.000	[-0.001, 0.000]	-0.000
b2	21	0.001	[-0.001, 0.001]	0.000
c2	21	-0.000	[-0.001, -0.000]	-0.000

S4D minus Mamba

Measure	Recordings	Mean difference	Exploratory interval	Equal-work difference
Full	21	0.020	[0.004, 0.034]	0.018
Active	21	0.015	[-0.004, 0.031]	0.011
Note	21	0.026	[-0.029, 0.066]	0.012
Within	21	0.017	[0.005, 0.028]	0.016
Error	21	0.115	[-0.070, 0.226]	0.062
b2	21	0.216	[-0.159, 0.430]	0.110
c2	21	-0.019	[-0.028, -0.009]	-0.017

Work-specific paired effects. The compositions, not movements, define the dependence groups. No composition is selected for a favorable result. **Mamba minus Instrument OLS**

Work	Full	Note	Error	b^2	c^2
BWV1001	0.076	0.228	-0.102	-0.118	-0.014
BWV1002	0.006	0.111	-0.025	-0.089	0.022
BWV1006	0.065	0.304	-0.043	-0.074	-0.035
BWV1009	0.152	0.144	-0.027	0.082	-0.139
BWV1010	0.123	0.100	-0.167	-0.188	-0.166

Mamba minus Shared envelope

Work	Full	Note	Error	b^2	c^2
BWV1001	0.040	0.029	-0.116	-0.182	0.015
BWV1002	-0.080	-0.235	-0.060	-0.234	0.053
BWV1006	-0.010	-0.032	-0.064	-0.077	-0.088
BWV1009	0.071	0.145	-0.260	-0.460	-0.002
BWV1010	0.068	0.104	-0.286	-0.562	-0.004

S4D minus Instrument OLS

Work	Full	Note	Error	b^2	c^2
BWV1001	0.099	0.305	-0.136	-0.123	-0.048
BWV1002	0.031	0.050	-0.140	-0.411	0.000
BWV1006	0.064	0.268	-0.081	-0.142	-0.034
BWV1009	0.153	0.152	0.198	0.483	-0.149
BWV1010	0.164	0.175	0.104	0.353	-0.188

S4D minus Shared envelope

Work	Full	Note	Error	b^2	c^2
BWV1001	0.063	0.106	-0.150	-0.187	-0.020
BWV1002	-0.055	-0.296	-0.175	-0.556	0.032
BWV1006	-0.012	-0.069	-0.102	-0.144	-0.086
BWV1009	0.072	0.152	-0.034	-0.058	-0.012
BWV1010	0.109	0.179	-0.015	-0.021	-0.026

Shared envelope minus Note OLS

Work	Full	Note	Error	b^2	c^2
BWV1001	-0.007	0.001	-0.001	-0.000	-0.000
BWV1002	0.030	-0.000	-0.000	-0.000	0.000
BWV1006	-0.015	0.001	-0.001	-0.002	-0.000
BWV1009	-0.013	0.005	0.000	0.001	-0.001
BWV1010	-0.012	0.001	0.001	0.002	-0.000

S4D minus Mamba

Work	Full	Note	Error	b^2	c^2
BWV1001	0.023	0.077	-0.034	-0.005	-0.035
BWV1002	0.025	-0.061	-0.115	-0.322	-0.022
BWV1006	-0.002	-0.037	-0.038	-0.067	0.001
BWV1009	0.001	0.008	0.225	0.402	-0.010
BWV1010	0.041	0.075	0.271	0.540	-0.022

U. Fixed-output scale views and reproducibility

These additional views use the same saved outputs. They do not select models, recordings or thresholds. Log uses epsilon 10^{-7} before Gaussian smoothing; the Gaussian support is four standard deviations with reflection padding, matching the original analysis. The following equal-recording means use log-amplitude at zero, 0.05, 0.1, 0.2, 0.5 and 1-second smoothing. All 21 correlations are defined in each cell. Complete linear-smoothed values are in the machine-readable report.

Predictor	0	.05 s	.1 s	.2 s	.5 s	1 s
Gate	0.491	0.613	0.642	0.672	0.711	0.736
Global OLS	0.460	0.587	0.618	0.650	0.693	0.720
Instrument OLS	0.412	0.529	0.559	0.587	0.628	0.653
Note OLS	0.492	0.611	0.640	0.671	0.710	0.734
Shared envelope	0.493	0.610	0.640	0.670	0.709	0.733
Mamba	0.532	0.607	0.633	0.660	0.699	0.728
S4D	0.521	0.624	0.658	0.687	0.729	0.755

Source and repeatability. MusicNet: John Thickstun, Zaid Harchaoui and Sham M. Kakade, *Learning Features of Music from Scratch*, ICLR 2017. Official raw release: <https://zenodo.org/records/5120004>. The accompanying MusicNet analysis directory includes the frozen plan, derived features/targets, all seven output arrays, metadata, hashes, work effects and a CPU metric-reproduction program. It includes no recordings. Source acquisition checksums and exact preprocessing instructions enable a separately licensed source download.

The frozen plan predates new outputs; runtime and annotation-handling amendments are timestamped. The original 186-track results, seeds and historic test exposure remain separate. Input, checkpoint and source hashes, environment and local/remote execution identities remain in the provenance records. This new test neither converts the old comparisons into clean confirmatory experiments nor establishes that every diagnostic effect generalizes.

V. Direct verification of shared-template mean preservation

This retrospective check was specified after reviews of v0.37. It uses every saved seed-42 Note OLS and shared-envelope prediction, with no fitting or output modification. For each eligible original note, compare its predicted interval means before/after shape rendering. Relative difference divides by the larger of the Note OLS mean and 10^{-7} . Log difference uses the original 10^{-7} floor. Direct averaging is independently checked against cumulative-sum averaging.

Frame-nonoverlap means that no frame in the eligible interval is shared with any other annotated note, including notes shorter than four frames. This grid-based check retains 29,769 notes, unlike the earlier continuous-time overlap exclusion (28,077); they answer different boundary questions and are not substituted for one another. The maximum all-note relative change of 16.5% occurs at an overlap, so the control is not asserted to preserve every overlapping interval exactly.

Population	Notes	Median relative change	95th percentile	Maximum
All original notes	30352	3.1e-09	1.2e-08	0.165
No shared frames	29769	3.02e-09	1.06e-08	3.56e-08

Across 186 tracks	Median absolute change	95th percentile	Maximum
Note correlation	3.04e-09	0.00112	0.0085
Note log-RMSE	4.25e-10	0.000197	0.00837

Complete-horizon, nonoverlapping-track comparison. All 133 tracks with no shared annotated frames enter this subset. Their original complete horizons and silent gaps are retained. Full is not recomputed on a masked note union. This subset checks the control without overlap mixing; it is not a replacement for the original 186-track population.

Predictor	Full	Note	Error
Note OLS	0.467336024	0.196540866	0.741846858
Shared envelope	0.546033345	0.196540866	0.741846857

On this subset, adding shared shapes increases mean Full by 0.0787 while Note and Error are unchanged to numerical precision. Across every frame-nonoverlap note, relative interval-mean change is at most 3.56×10^{-8} . Thus the shape-control result does not require overlap-induced changes of interval means. The full per-track report, frozen audit plan and CPU replay program accompany the analysis arrays. Cumulative-sum verification differs from direct means by at most 1.42×10^{-15} .

W. Transfer shape coverage and overlap check

This follow-up uses the unchanged MusicNet outputs after round-four review. Its separately timestamped plan is retrospective and does not extend the original pre-outcome selection protocol. All 21 recordings and every predictor are retained. Among 19,713 annotations, 19,699 have at least four frames; 17,014 eligible intervals share no frame with another annotation. All annotations, including short notes, enter the overlap count.

The table reports original and frame-nonoverlap Within coverage. Nonoverlap means first average defined intervals within each recording, then average recordings equally. A constant Note OLS interval has undefined Within; its 639 originally defined intervals disappear entirely after overlap exclusion. Thus its original .101 over only 14 recordings cannot be ranked against the other predictors' 21-recording shape results.

Predictor	Original defined notes	Nonoverlap defined notes	Recordings	Within
Gate	0	0	0	–
Global OLS	19699	17014	21	0.034
Instrument OLS	19699	17014	21	0.006
Note OLS	639	0	0	–
Shared envelope	19699	17014	21	0.014
Mamba	19699	17014	21	0.054
S4D	19699	17014	21	0.078

The shared template has original Within .026, falling to .014 on nonoverlap intervals. Mamba and S4D give .054 and .078 on the same 17,014 nonoverlap intervals. Shape transfer is weak even without overlapping annotations; this does not identify its acoustic cause. Note and Error for the unshaped/shaped references match to numerical precision on those intervals (Note .014987140/.014987139; Error 1.063678748/1.063678748).

Three complete recordings have no shared annotated frames: 2288, 2298 and 2659. Their original complete horizons, including gaps, give Full .199615 for Note OLS and .191942 for shared envelopes. Full is not computed on a concatenated mask. The absence of an average template gain therefore also occurs in this small nonoverlap subset; it is not evidence that overlap alone caused the full-population failure. All per-recording values, missing cases, input hashes and the CPU replay script are supplied.

X. Duration-weighted note correlation on nonoverlapping tracks

This retrospective sensitivity check was specified after round-five review of v0.39. It retains all 133 original seed-42 tracks with no shared annotated frame, their complete horizons and unchanged predictions. Full remains the original full-track correlation, including silent gaps. Note and duration-weighted Note use identical eligible intervals and identical interval means. Only their within-track weights change: uniform weights or interval frame counts normalized to sum to one. Weighted Pearson centers both note-mean vectors by their weighted means, then divides weighted covariance by the product of weighted standard deviations.

The denominator below is the number of tracks with a positive original Full difference. Uniform/duration columns count negative Note differences under the respective rule; both counts their intersection. These are descriptive fixed-output counts, not independent new observations or a causal decomposition of Full. Across-track means weight tracks equally under both note rules.

Model / reference	Full up	Uniform down	Duration down	Both down
Mamba / Instrument OLS	106	35	27	21
Mamba / Shared envelope	108	16	15	10
S4D / Instrument OLS	112	35	36	25
S4D / Shared envelope	108	23	12	9
BiGRU / Instrument OLS	107	55	55	45
BiGRU / Shared envelope	99	29	27	23

Model / reference	Mean Note difference	Mean duration-weighted difference
Mamba / Instrument OLS	0.023	0.041
Mamba / Shared envelope	0.095	0.117
S4D / Instrument OLS	0.002	0.027
S4D / Shared envelope	0.075	0.104
BiGRU / Instrument OLS	-0.037	-0.035
BiGRU / Shared envelope	0.036	0.041

For Mamba versus instrument OLS, disagreement falls from 35/106 to 27/106, but only 21 tracks disagree under both rules: 14 cease to disagree and six newly disagree. S4D changes from 35/112 to 36/112, whereas BiGRU remains 55/107 with changed membership. Thus changing note-duration weights changes both counts and track identity; it does not uniformly remove disagreements. Duration-weighted Note still omits within-note shape and silent-gap behavior. These counts cannot assign each neural-model reversal to shape, duration or silence separately. The distinct shared-template intervention remains the direct test of changing shape while holding nonoverlapping interval means fixed.

Y. Core-control uncertainty and transfer disagreement magnitudes

These requested descriptive clarifications use unchanged outputs after round-six review. The no-overlap subset contains 133 complete tracks in 34 piece/duplicate groups. Resample those 34 whole groups 5,000 times with default RNG seed 20260911, retaining every sampled track and recomputing the equal-track mean Full difference. Shared envelope minus Note OLS is .078697, with percentile interval [.058547,.102878]. This is conditional, retrospective uncertainty for the selected annotation-defined population, not a universal template benefit. Its 93 URMP tracks average +.050098 and its 40 Bach10 tracks +.145191; PHENICX and TRIOS contribute no completely nonoverlapping tracks to this subset.

Every MusicNet Full-up/Note-down case against instrument OLS appears below. Positive Full difference and negative Note difference define disagreement; magnitude was not used to select recordings for the original evaluation. Applying the already reported double .05 threshold leaves 1/14 for Mamba and 1/18 for S4D. Denominators are recordings with Full gain above .05, not all recordings or five independent works. This shows numerical disagreements remain under the stricter rule, not that their original rate transfers.

Predictor	Recording	Full difference	Note difference
Mamba	2220	0.015224	-0.050294
Mamba	2221	0.054771	-0.079247
Mamba	2294	0.033425	-0.090470
S4D	2221	0.059828	-0.057522
S4D	2294	0.062951	-0.032212

Plans, full-precision values and CPU scripts for sections V–Y accompany the saved arrays. These later audits are explicitly retrospective. They do not change original outputs, fitting roles, selection rules or the outcome-frozen MusicNet comparison.

Z. Fixed-output support and file-end sensitivity

This retrospective check applies the timestamped round-seven plan to the unchanged original 186-track arrays. Integer-truncated, horizon-clipped note intervals contain at least four frames; Pearson correlation needs three samples and standard deviations of at least 10^{-10} . All 30,352 eligible intervals enter the original Note/Error calculation. Each fitted reference and neural predictor has defined Note and Error on all 186 tracks. The unit gate has undefined Note on every track because its eligible note means are constant; its Error is defined on all 186. No undefined correlation is replaced by zero.

Error decomposition. For each track, let $d_j = \log(\hat{a}_j + 10^{-7}) - \log(a_j + 10^{-7})$ across its eligible interval means. The audit recomputes Error, $b^2 = (\text{mean}_j d_j)^2$ and $c^2 = \text{mean}_j (d_j - \text{mean}_j d_j)^2$. The following are equal-track means; squaring the displayed mean Error does not give mean $b^2 + c^2$.

Predictor	Note support	Error	b^2	c^2
Gate	0	3.708426	14.075691	0.202902
Global OLS	186	0.814200	0.598623	0.216121
Instrument OLS	186	0.745229	0.453728	0.228795
Note OLS	186	0.723253	0.428706	0.206326
Shared envelope	186	0.723343	0.428789	0.206427
Mamba	186	0.688952	0.397147	0.194580
S4D	186	0.718275	0.474681	0.192456
BiGRU	186	0.649831	0.282251	0.210666

Endpoint guard. Full excludes the first two and last two cached RMS frames of every track, comparing prediction and target on $[2 : -2]$. Note excludes an entire eligible interval whenever it intersects those margins: retain only integer endpoints $s \geq 2$, $e \leq T - 2$. All 186 tracks remain; 30,312 of 30,352 intervals remain. All retained interval means use their original frames, with no shortening or new target extraction. The rule is identical across tracks and predictors.

For a centered 320-sample RMS window and 160-sample hop, file-end padding can affect endpoint windows. This guard tests whether the reported comparisons rely on those boundary frames. It does not identify the historical padding mode, re-extract RMS under zero/reflection alternatives, test other preprocessing choices, or shift annotation boundaries.

Predictor / reference	Frames	Full difference	Note difference	Discordant / Full up
Mamba / Instrument OLS	Original	+0.106261	+0.030206	50/152
Mamba / Instrument OLS	Guarded	+0.106406	+0.031031	50/152
Shared envelope / Note OLS	Original	+0.073257	-0.000092	80/163
Shared envelope / Note OLS	Guarded	+0.073580	-0.000092	80/163

Mamba versus instrument OLS retains 50/152 Full-up/Note-down tracks. Shared envelope versus Note OLS retains a mean Full gain of .073580, with mean Note change -.000092. Its sign-only discordance count includes minute numerical or overlap-related interval-mean changes and should be read alongside the direct preservation audit in section V. These fixed-output results show that deleting file-end margins leaves the two reported patterns intact under the stated rule.

Illustration and replay. Figure 2 retains the previously chosen discordant case nearest the quadrant median after IQR scaling (ties by track ID), and the same central ten-second window as section B. It displays every eligible whole-track note mean and all 186 paired track differences. The illustrative selection followed aggregate inspection; neither case nor window was reselected for this audit. The package includes the figure generator and input arrays, the round-seven plan, every per-track result, and an independent CPU verifier using cumulative sums for interval means and centered dot products for correlations.